

Contestant Number: _____

Time: _____

Rank: _____



Managerial Accounting

(135)

STATE 2025

CONCEPT KNOWLEDGE:

Multiple Choice (15 @ 2 points each) _____ (30 points)

APPLICATION KNOWLEDGE:

Short Answer _____ (33 points)

Problem 1: CVP Analysis _____ (30 points)

Problem 2: Variable Costing _____ (30 points)

TOTAL POINTS _____ ***(123 points)***

Test Time: 90 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

EXAM GUIDELINES:

You have been hired as a Financial Assistant and will be keeping the accounting records for Digital Solutions, located at 700 Morse Road, Suite 201, Columbus, Ohio 43214. Digital Solutions provides accounting and other financial services for clients. You will complete jobs for Digital Solutions' own accounting records, as well as for clients.

You will have 90 minutes to complete your work. The test is divided into two parts: concept knowledge and application knowledge.

Your name and/or school name should *not* appear on any work you submit for grading. Write your Member ID in the provided space. Staple all pages in order before you turn in your test.

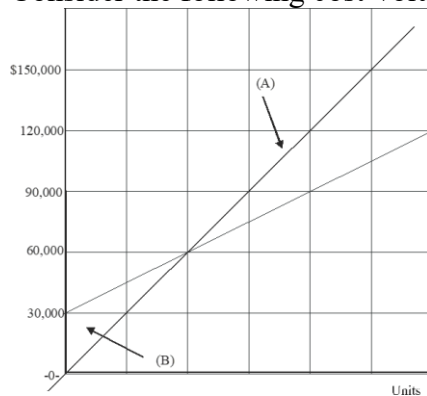
Assumptions to make when taking this assessment:

- Round all calculations to two decimals at the final step.
- Round all percentages to one decimal place.
- Use 360 days for interest calculations.

Multiple Choice Questions

Directions: Identify the letter of the choice that best completes the statement or answers the question.

- Celtics Company pays its production managers a salary of \$5,000 a month. Salespersons, however, are paid strictly on commission at \$2.00 for each case of product sold. Production managers' salaries are an example of which of the following cost behaviors?
 - Variable
 - Fixed
 - Mixed
 - Curvilinear
- Once sales reach breakeven point, then each additional unit sold will _____.
 - Increase fixed costs by a proportionate amount
 - Reduce margin of safety
 - Increase profit by an amount equal to contribution margin per unit
 - Increase the company's operating leverage
- Fixed cost per unit _____.
 - Decreases as production volume decreases
 - Is not affected by changes in volume
 - Increases as production volume increases
 - Decreases as production volume increases
- Consider the following cost-volume-profit graph:



What is the approximate amount of fixed costs in this organization?

- \$0
- \$25,000
- \$30,000
- \$60,000

5. Knicks, Inc. normally produces between 50,000 and 75,000 units each year. Producing more than 75,000 units alters the company's cost structure. For example, fixed costs increase because more space must be rented, and additional supervisors must be hired. The production range between 50,000 and 75,000 is called the ____.
- A. Differential range
B. Relevant range
C. Operational range
D. Leverage range
6. A product has a contribution margin of \$3 per unit and selling price of \$10 per unit. Fixed costs are \$18,000. Assuming new technology doubles the unit contribution margin but increases total fixed costs by \$12,000, what is the breakeven point in units?

- A. 2,000 units
B. 3,000 units
C. 5,000 units
D. 6,000 units

7. Which of the following costs is a mixed cost?

Units Sold	20	40	60
Total salary cost	\$1,200	\$1,600	\$2,400
Total cost of goods sold	\$1,800	\$3,600	\$5,400
Total rent cost	\$3,200	\$3,200	\$3,200

- A. Total salary cost
B. Total cost of goods sold
C. Total rent cost
D. All costs are mixed
8. At the break-even point, ____.
- A. Fixed costs equal variable costs
B. Total sales equal fixed costs
C. Total sales equal total costs
D. Total sales equal variable costs
9. When total fixed costs increase, the break-even point ____.
- A. Decreases
B. Remains the same
C. Increases
D. Decreases proportionately
10. When comparing a traditional (GAAP) income statement to a CVP income statement:
- A. Net income will always be greater on the traditional statement
B. Net income will always be less on the traditional statement
C. Net income will always be identical on both when inventory stays the same.
D. Net income will be greater or less depending on the sales volume

11. Hornets Inc. has expected sales of \$12,000 and break-even sales of \$8,400. What is the margin of safety in dollars?
- A. \$1,200
 - B. \$840
 - C. \$3,600
 - D. \$3,000
12. Wizard's Bookstore has an ROI of 15%. Assuming both sales and expenses cannot be changed, what can Wizard's do to increase ROI to 21%?
- A. Increase operating assets
 - B. Decrease operating assets
 - C. Increase liabilities
 - D. Nothing can be done to change ROI
13. Which of the following best describes a standard?
- A. Budgeted amount for the total product cost
 - B. Cost or quantity that is expected under normal conditions
 - C. Costs incurred to produce a product
 - D. Actual sales price, cost, or quantity
14. A 15% increase in production volume will result in a ____.
- A. 15% increase in total variable costs
 - B. 15% increase in total fixed costs
 - C. 15% increase in variable costs per unit
 - D. 15% increase in total mixed costs
15. Which is NOT one of the four most commonly used perspectives on a balanced scorecard?
- A. The financial perspective
 - B. The customer perspective
 - C. The learning & growth perspective
 - D. The external process perspective

Short Answer (33 Points)

1. The following income statement is provided for Timberwolves Co. Timberwolves produced and sold 2,400 units in March.

Sales revenue	\$50,000
Cost of goods sold (variable)	20,000
Administrative salaries	6,000
Depreciation	4,000
Office Supplies(variable)	4,800

What is the company's contribution margin for March?	
What is the company's contribution margin per unit?	

2. Raptors is an Internet service provider that is particularly popular with college students. The company is interested in estimating the fixed and variable components of its technical support costs. The manager believes that these costs are driven by the number of hours of support given. The following information was gathered for the last six months of business:

<u>MONTH</u>	<u>NUMBER OF HOURS</u>	<u>TECHNICAL SUPPORT COSTS</u>
January	4,300	\$125,000
February	2,150	\$ 75,000
March	1,800	\$ 75,000
April	3,600	\$100,000
May	1,500	\$ 55,000
June	2,900	\$ 90,000

Use the high-low method to estimate the variable cost per hour.	
Use the high-low method to estimate the total fixed cost.	
If Raptors provides 3,200 hours of technical support, what are the estimated technical support costs?	

3. Clippers Corp. has expected sales ranging from \$170,000 to \$200,000. Variable costs and their percentage relationships are: Sales commissions 5%, Advertising 4%, Traveling 3%, and Supplies 8%. Fixed expenses are Salaries \$34,000, Depreciation \$7,000, and Insurance \$1,000.

If sales are \$180,000 what are the total variable costs expected to be?	
If sales are \$180,000 what are the total fixed costs expected to be?	
If sales are \$180,000 what is profit expected to be?	

4. Thunder Corporation produces and sells a single product for \$140. The variable costs are \$84 per unit. The company is currently selling 5300 units per month. Fixed expenses are \$185,000 per month. The marketing manager believes that a \$5300 increase in the monthly advertising budget would result in a 200 unit increase in monthly sales.

What would Thunder's net income be after this change?	
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5. Pacers Inc. currently has sales of \$273,600, a contribution margin of \$150,480, and net operating income of \$100,820.

What is the Pacers' degree of operating leverage?	
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How much Pacers net income increase by if sales increase by 15%?	
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Problem 1: Cost-Volume Profit Analysis (30 Points)

Pacers Company has provided the following information

Sales price per unit	\$50
Variable cost per unit	\$18
Fixed costs per month	\$17,000

Calculate the following:

Contribution margin per unit	
Contribution margin ratio	
Required sales in units to break-even	
Sales revenue required to break-even	
Required sales in units to earn \$5,000 profit	
Sales revenue required to earn \$5,000 of profit	

Problem 2: Variable Costing (30 Points)

Pelicans Corp. incurs the following costs for their famous bowties during the first quarter of 2025.

Direct materials per tie	\$9
Direct labor per tie	\$5
Variable manufacturing overhead per tie	\$3
Total fixed manufacturing overhead	\$60,000
Total fixed selling and administrative expenses	\$25,000
Sales price per tie	\$49
Bowties produced	10,000
Bowties sold	8,000

What is the product cost per tie under absorption costing?	
What is the product cost per tie under variable costing?	
What is gross profit under absorption costing?	
What is the contribution margin under variable costing?	
What is net income under absorption costing?	
What is net income under variable costing?	